

CS340 (2025) – Quiz 1

Duration: 40 minutes, Total marks: 20, Pages: 5

- **Important note.** Answers without clear and concise explanations will not be graded.

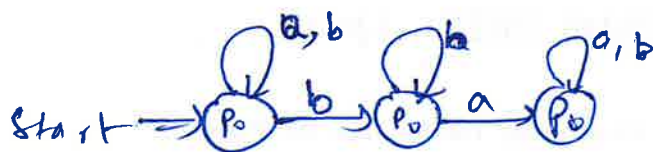
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Problems

1. (4 marks) Let $\Sigma = \{0, 1\}$ and $A = \{x \in \Sigma^* \mid x \text{ does not represent a multiple of 3 in binary}\}$.
(Note: leading zeros are permitted in the input string, the empty string ϵ represents the number 0.)
Construct a DFA M with at most 3 states such that $L(M) = A$. Clearly justify your answer.

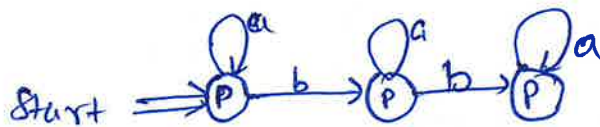
2. (4 marks) Let $A = \{x \in \{a, b\}^* \mid \text{every 'a' in } x \text{ is never followed by 'bb'}\}$. Construct a DFA M with at most 4 states such that $L(M) = A$. Clearly justify your answer.



3. (4 marks)

Let $A = \{x \in \{a, b\}^* \mid x \text{ contains at least two 'a's'}\} \cup \{x \in \{a, b\}^* \mid x \text{ contains exactly two 'b's'}\}$.
 Construct a DFA or NFA M with at most 6 states such that $L(M) = A$. Clearly justify your answer.

$$\Sigma = \{a, b\}^* \mid x \text{ contains at least two 'a's'} \cup \{x \in \{a, b\}^* \mid x \text{ contains exactly two 'b's'}\}$$



- $$\begin{array}{ccc} a & & a, b \end{array}$$

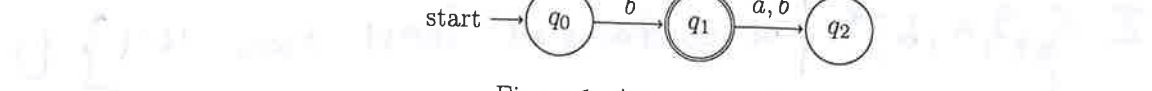


Figure 1: Automaton M_1

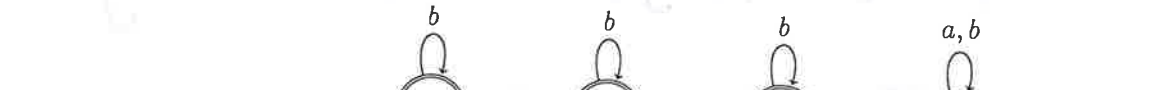


Figure 2: Automaton M_2

$L(M_1) = \{x \in \{a,b\}^* \mid \text{starts with 'ab'}\}$

$L(M_1) = \{x \in \{a,b\}^* \mid \text{starts with 'bababa'}\}$

5. (4 marks) Consider the automaton M_3 given in Figure 3. Is $L(M_3)$ finite? Clearly justify your answer.

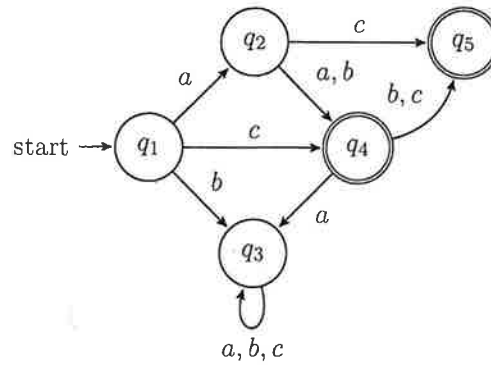


Figure 3: Automaton M_3

Rough

~~Quelent~~
~~too much~~
①②③

113

100

101

1106

1100

10019